

The Scope For **BLOCKCHAIN** ADOPTION In The **ENERGY** SECTOR

As everyone jumps onto the blockchain bandwagon to implement it in sectors like finance and land records, its use in the energy distribution sector seems to hold a lot of promise. This article explores its potential



After Bitcoin demonstrated the potential of blockchain, people from many industries sat up. Though the financial services sector was an early adopter, it did not use this technology merely for cryptocurrency purposes, but rather for reasons of security and authentication. A publicly distributed chain of blocks, each block representing data or a transaction, was virtually tamper-proof. Soon, other verticals, like the energy sector, began exploring the potential of blockchain.

The Brooklyn Grid

One of the early proofs of concept (PoCs) of how blockchain could work in the utilities space is the Brooklyn Grid that was set up in 2015. This began in a single neighbourhood in New York. Some residents set up solar panels that fed the grid—these are whom we call prosumers. Others were just consumers. This part of the operations involved Siemens (multinational in the electricity space, which took care of digital grids and components) and LO3 Energy (firm that develops blockchain-based solutions that have changed how energy is generated, stored, bought and sold, at a local level).

At the first stage, with neighbours on the grid connected to each other, the system worked as follows:

1. A home solar energy generator's units of power were tokenised as it got fed into the common grid.

2. Tokens could be bought directly by others in the grid, who were just electricity consumers, bypassing the need to deal with a centralised authority or utility.

3. Transactions were stored in a public Ethereum blockchain.

In his talk about blockchain usage in the energy sector, at *India Electronics Week (IEW)* in February this year, Prashant K. Singh, senior consultant, Blockchain Centre of Excellence, Wipro, described how things work. This peer-to-peer trading in electricity tokens was enabled by an app that offered two views: one of the users' own consumption, billing and trading history, as well as that of the neighbours. This allowed residents to compare their own energy usage with that of other residents and maybe change behaviour to ensure optimised consumption. It also let them know which homes had tokens available for peer-to-peer trading.

In 2016, this pilot project got upgraded further by the installation of smart meters by blockchain middleware provider ConsenSys, and the incorporation of many aspects of the Internet of Things (IoT). The software on these meters enable smart transactions, whereby energy usage is tracked automatically, bills